



Classes and Methods

Practical Skills Module 4: Topic 1

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Topic Code	PM-04-PS01
Topic	Define classes, functions or methods

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0101 Use the keyword to define a class
- PA0102 Define functions or methods within the class
- PA0103 Use indentation
- PA0104 Make an object of the class
- PA0105 Call a method in a class
- PA0106 Demonstrate use of inheritance

Applied Knowledge

- AK0101 Concept, definition and functions of intermediate Python programming principles

Internal Assessment Criteria

- IAC0101 Expected results with Python OOP are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-04



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Employee Management System

A company wants to create an **Employee Management System** using Python to manage employee details. The system will define a class for employees, include methods for tasks like displaying details, and demonstrate inheritance by creating specialized classes for different types of employees (e.g., Manager, Developer).

Create a Python file called **employee_management_system.py** and complete the following steps:

Step 1: Create a class to represent a generic employee.

- Use the `class` keyword to define a class called `Employee`.
- Include a constructor (`__init__` method) that sets up at least two attributes:
 - `name` (the employee's full name)
 - `employee_id` (a unique ID for the employee)

Expected input example (values used when creating an object):

```
Name: "Alice Johnson"  
Employee ID: "E101"
```

Step 2: Add a Method to Display Details

Inside the `Employee` class, define a method that prints the employee's name and employee ID in a readable format.

Expected output format:

```
Employee Name: Alice Johnson  
Employee ID: E101
```

Step 3: Ensure Correct Indentation

Ensure your constructor and method(s) are indented correctly inside the class. Python relies on indentation to group code blocks.

Step 4: Create an Instance of the Class

Below the class definition, create an **object** of the `Employee` class using sample data (e.g., a name and employee ID of your choice).

Step 5: Call the Method Using the Object

Use the object to call the method you defined in Step 2. This should display the employee's name and ID.

Expected output when calling the method:

```
Employee Name: Alice Johnson  
Employee ID: E101
```

Step 6: Demonstrate Inheritance

Create two new classes that inherit from the `Employee` class:

- `Manager`
- `Developer`
 - Each class must use inheritance.
 - Add one extra attribute for each subclass:
 - For `Manager`, this could be `department`.
 - For `Developer`, this could be `programming_language`.
 - Each subclass should override or extend the method to display all relevant details.

Step 7: Create and Test Subclass Objects

Create one object from each subclass (`Manager` and `Developer`) using your own sample data. Then, call the display method for each to show the full employee details.

Example expected outputs:

```
Employee Name: Bob Smith  
Employee ID: M202  
Department: HR
```

Employee Name: Clara Davis
Employee ID: D303
Programming Language: Python



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
